School of Future Tech

Case Study Report

on

Online Quiz Tournament Platform

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1. Introduction to the Case Study

Online quiz platforms are widely used in education, learning competitions, and knowledge testing. However, most quiz systems focus on simple single-player quizzes and do not provide a structured tournament environment where players compete through multiple rounds and track rankings.

This case study focuses on designing and implementing an **Online Quiz Tournament Platform** using **HTML5, CSS, and JavaScript**. The platform allows users to participate in a multi-round quiz competition where players answer timed multiple-choice questions and advance through rounds based on their scores.

The system includes features such as a **timer-based quiz, score calculation, round progression, and a leaderboard to display rankings**. All functionality runs entirely in the browser using frontend technologies, without requiring a backend server.

This project demonstrates how fundamental web technologies and client-side scripting can be used to build an interactive quiz platform that simulates a real tournament-style learning environment.

2. Problem Statement / Case Background (Abstract)

Background

Most online quiz platforms provide simple question-answer formats for individual users. They often lack a structured competitive environment where players can progress through multiple rounds, compete against rankings, and track performance across a tournament. This limits engagement and reduces the competitive learning experience.

Abstract

This case study presents the design and implementation of an **Online Quiz Tournament Platform** built using **HTML5, CSS, and JavaScript**. The system allows users to participate in a structured quiz competition consisting of multiple rounds such as Round 1, Quarterfinal, Semifinal, and Final.

Players answer timed multiple-choice questions, and scores are calculated based on correctness and response time. The platform includes features such as question display, countdown timers, score tracking, round progression, and a leaderboard to display rankings.

All data such as scores and player rankings are stored using **localStorage**, allowing the platform to function entirely on the frontend without requiring a backend server. The project demonstrates how core web technologies can be used to build an interactive and competitive quiz application.

3. Case Study Design

The **Online Quiz Tournament Platform** system is designed as a modular pipeline with the following stages:

1. The **Online Quiz Tournament Platform** is designed as a structured frontend system with the following stages:
2. **User Registration**
   Players enter their name and receive a unique Player ID. The information is stored using **localStorage** so returning players can continue their progress.
3. **Quiz Initialization**
   The system loads quiz questions from a question bank stored in a JSON file. Questions are randomly selected for each tournament session.
4. **Round-Based Tournament Flow**
   The quiz follows a structured tournament format:
   Round 1 → Quarterfinal → Semifinal → Final.
   Players must achieve a minimum score in each round to advance.
5. **Question Display and Timer**
   Each question is displayed with four options and a countdown timer. Players must select the correct answer before the timer expires.
6. **Score Calculation**
   Scores are calculated based on correct answers and response time. Faster responses receive higher points.
7. **Leaderboard and Result Tracking**
   At the end of the tournament, scores are stored in **localStorage** and displayed on a leaderboard showing player rankings and top performers.
8.

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Key methods and algorithms:

1. **Question Randomization**
   Quiz questions are stored in a JSON question bank and shuffled before the tournament begins. Random selection ensures that each tournament session presents questions in a different order.
2. **Timer Algorithm**
   Each question has a **10-second countdown timer** implemented using JavaScript's *setInterval()* function. If the timer reaches zero, the system automatically moves to the next question.
3. **Score Calculation Logic**
   Players earn points for correct answers. The score is influenced by response speed, rewarding faster answers with higher points while slower responses receive fewer points.

4. **Round Progression System**
   The tournament follows a structured flow:
   Round 1 → Quarterfinal → Semifinal → Final.
   Players must achieve a minimum score threshold in each round to advance.
5. **Leaderboard Storage**
   Player scores and rankings are stored using **localStorage** in the browser. This allows the leaderboard to persist between sessions without requiring a backend database.

Technology Stack Used for the Case

Frontend Technologies

- HTML5 for page structure and quiz interface
- CSS3 for layout, responsive design, and tournament UI styling
- JavaScript for quiz logic, timers, scoring system, and DOM manipulation

Data Storage

- **localStorage** is used to store player profiles, scores, and leaderboard rankings.

Data Format

- Quiz questions are stored in a **JSON file** and loaded dynamically using JavaScript.

Development Environment

- Developed using a code editor such as **VS Code** and tested in modern web browsers like Chrome or Edge.

5. Problem Statement / Case Study Implementation Details and Snapshots.

The **Online Quiz Tournament Platform** is implemented as a browser-based web application using **HTML, CSS, and JavaScript**. The project is organized into multiple folders and files that handle different parts of the system, such as player registration, quiz logic, leaderboard display, and forum interaction.

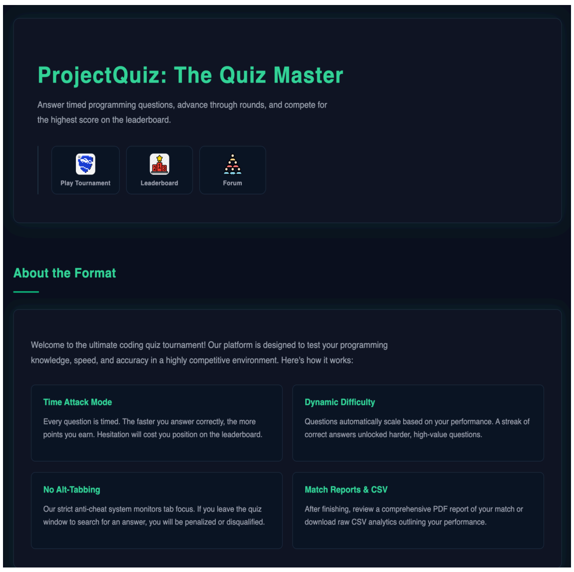
Project Structure

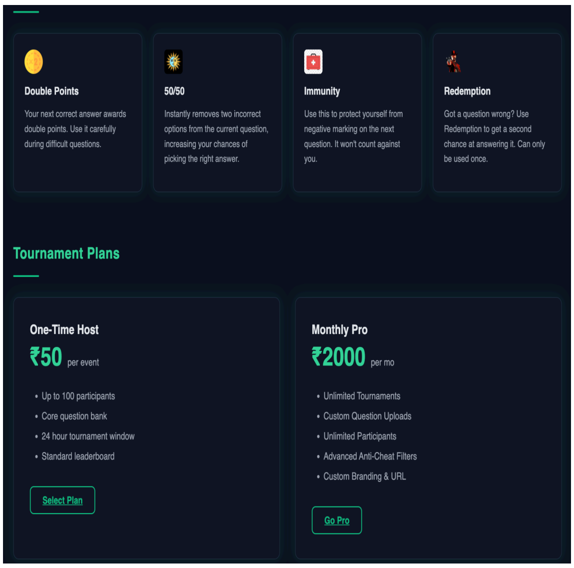
The application is divided into modules including:

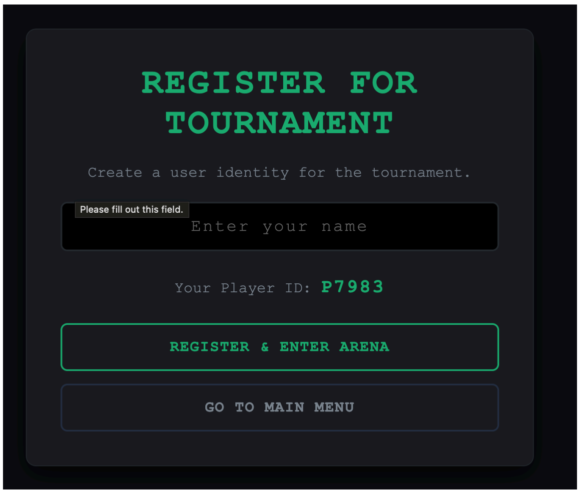
- **Landing Page (index.html)** – Introduces the platform and provides navigation to Play Tournament, Leaderboard, and Forum.
- **Registration Module** – Allows users to register with a name and generates a unique Player ID stored in localStorage.
- **Dashboard Module** – Displays player information such as name, ID, attempts used, and best score.
- **Quiz Engine** – Handles the tournament logic, including question display, timer countdown, answer validation, scoring, and round progression.
- **Leaderboard Module** – Shows ranked player scores stored in localStorage.
- **Forum Module** – Allows registered players to post comments and discuss the quiz.
- **Billing Module** – A simulated checkout page for premium tournament features.

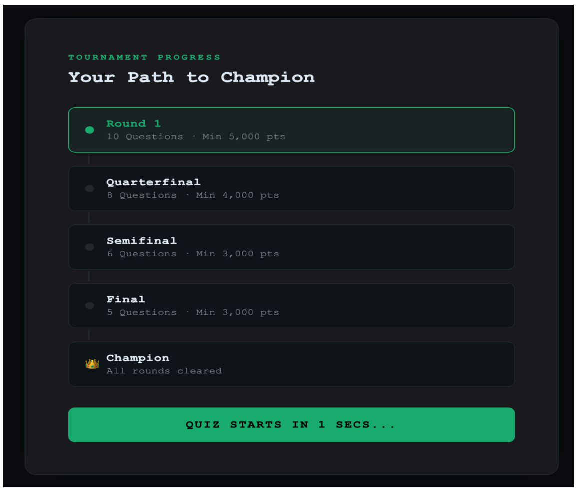
The system loads quiz questions from a **JSON question bank** and dynamically displays them using JavaScript. Player scores, leaderboard rankings, and comments are stored locally using **localStorage**, allowing the application to function without a backend server.

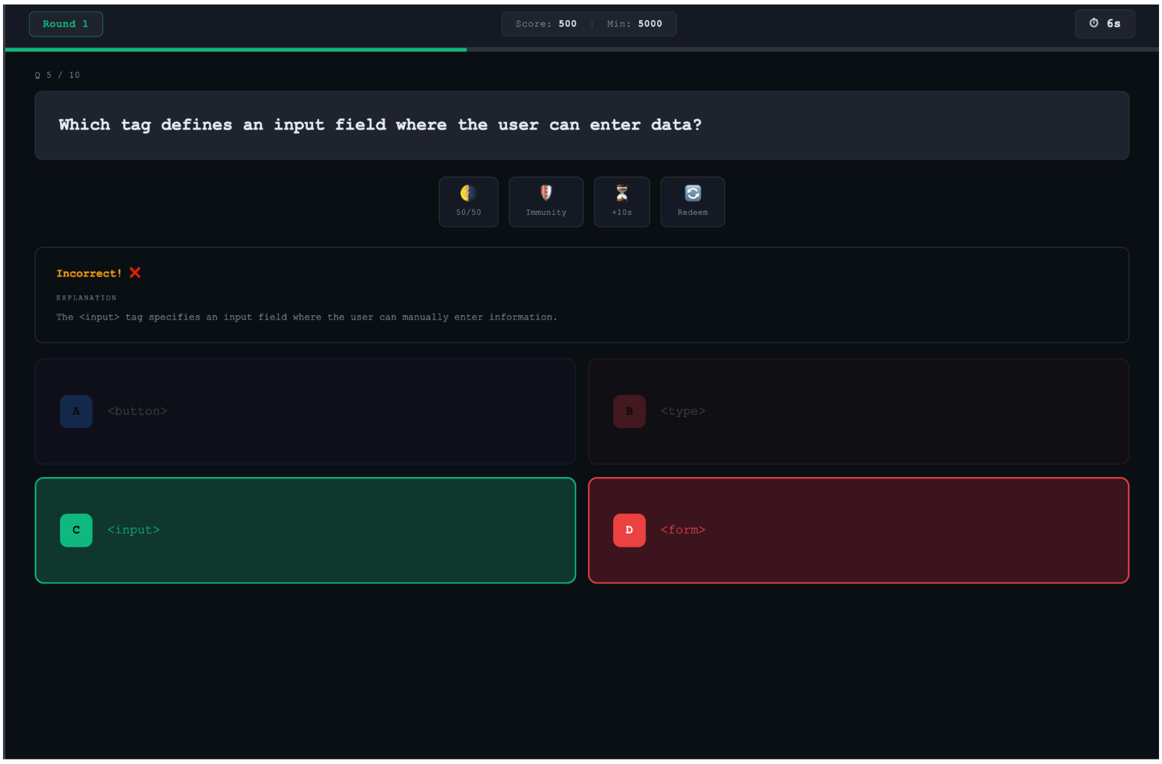
Snapshots to Include in the Report

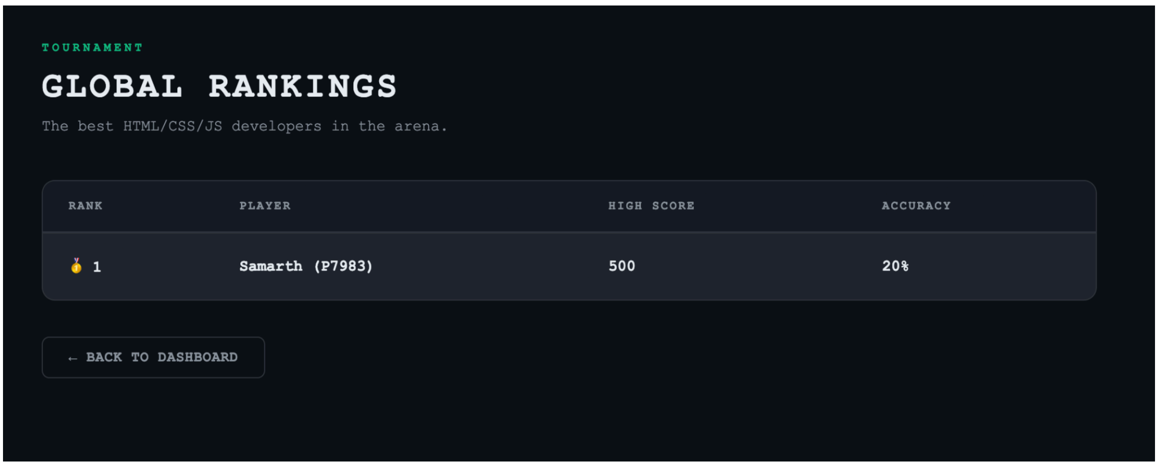


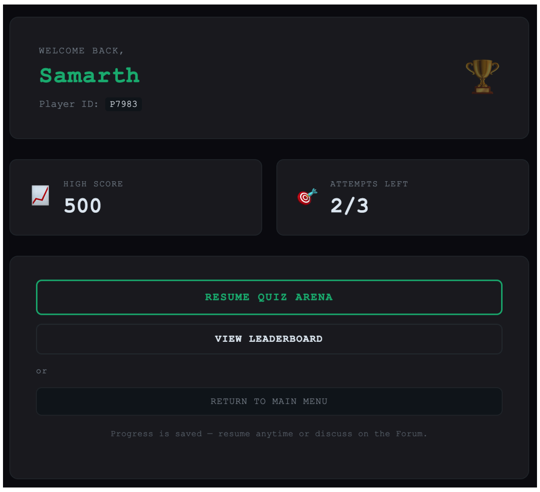


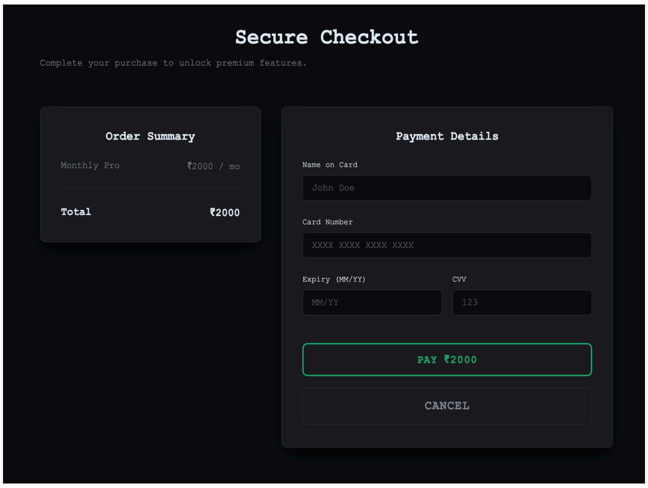

REGISTER FOR TOURNAMENT
Create a user identity for the tournament.
Please fill out this field.
Enter your name
Your Player ID: P7983
REGISTER & ENTER ARENA
GO TO MAIN MENU


TOURNAMENT PROGRESS
Your Path to Champion
Round 1
10 Questions · Min 5,000 pts
Quarterfinal
8 Questions · Min 4,000 pts
Semifinal
6 Questions · Min 3,000 pts
Final
5 Questions · Min 3,000 pts
Champion
All rounds cleared
QUIZ STARTS IN 1 SECS...


Round 1
Score: 500 | Min: 5000
6s
Q 5 / 10
Which tag defines an input field where the user can enter data?
50/50
Immunity
+10s
Redeem
Incorrect!
EXPLANATION
The <input> tag specifies an input field where the user can manually enter information.
A <button>
B <type>
C <input>
D <form>


TOURNAMENT
GLOBAL RANKINGS
The best HTML/CSS/JS developers in the arena.
RANK
PLAYER
HIGH SCORE
ACCURACY
1
Samarth (P7983)
500
20%
← BACK TO DASHBOARD





6. Problem Statement / Case Study Results and Conclusion.

Findings

The Online Quiz Tournament Platform successfully demonstrates that an interactive quiz competition system can be built using only frontend technologies. The platform manages player registration, timed quiz rounds, score calculation, and leaderboard ranking entirely within the browser.

The use of **JavaScript for quiz logic, timers, and scoring**, along with **localStorage for data persistence**, allows the system to function without a backend server. The round-based structure and leaderboard create a competitive environment that improves user engagement.

Conclusion

This case study demonstrates the design and implementation of an **Online Quiz Tournament Platform using HTML, CSS, and JavaScript**. The system integrates quiz gameplay, scoring logic, round progression, and leaderboard tracking into a single web application.

The project shows how core web technologies and client-side scripting can be used to build an interactive educational platform that simulates a competitive quiz tournament environment.

7. References

- Official documentation of **JavaScript (ES6)** for DOM manipulation, events, and client-side scripting.
- Documentation for the **Web Storage API (localStorage)** used for storing player data and leaderboard information.
- Online tutorials and educational resources on **building quiz applications using HTML, CSS, and JavaScript**.